Category Theory for Mathematical Physics — Exercise Sheet 2

Universal properties, adjunctions, semisimple categories

Variant: Hints and answers

Conventions as in Lectures 1 and 2: in $g \circ f$ the morphism f acts first; $\mathrm{Rep}(G)$ has the finite-dimensional complex representations; the simple objects of $\mathrm{Rep}(S_3)$ are $\mathbf{1}$, sgn and std (Example 2.15); $\mathbb{Z}$ and $\mathbb{Z}_n$ are written additively; $\zeta_n = e^{2\pi i/n}$. Characters and their inner product $\langle \chi, \chi' \rangle = \frac{1}{|G|} \sum_g \overline{\chi(g)} \chi'(g)$ are assumed from *Symmetries*.

Exercise 1 (Universal properties in the divisibility category). Let Div have objects $0, 1, 2, \ldots$ and exactly one morphism $m \to n$ when m divides n (so every m divides 0, and 0 divides only 0), and none otherwise.

(i) Show that Div is a category. Find its initial and its terminal object.

(ii) Show that the product of m and n is $\gcd(m, n)$ and the coproduct is $\mathrm{lcm}(m, n)$, with $\gcd(m, 0) = m$ and $\mathrm{lcm}(m, 0) = 0$.

(iii) Lemma 2.2 says a terminal object is unique up to a unique isomorphism. What are the isomorphisms of Div, and what does the lemma say here? In Set, show that the terminal objects are the one-point sets and the only initial object is $\emptyset$; in Vect, show that the zero space is both.

Hint. Divisibility is reflexive and transitive, and every Hom set has at most one element, so every diagram commutes. A product of m and n is a common divisor P that every common divisor divides.

Answer. (i) A category; 1 is initial and 0 terminal. (ii) gcd is the greatest common divisor in the order of divisibility, lcm the least common multiple. (iii) Only the identities: $m \mid n$ and $n \mid m$ force $m = n$, so a terminal object is unique on the nose. In Set the terminal objects are the one-point sets, unique up to a unique bijection, and $\emptyset$ is initial; in Vect the zero space is both.

Exercise 2 (Yoneda, in full). (i) Prove Theorem 2.6: for a functor $\Phi \colon C \to \mathrm{Set}$ and an object X, the natural transformations $\xi \colon \mathrm{Hom}(X, -) \Rightarrow \Phi$ correspond to the elements of $\Phi(X)$ by $\xi \mapsto \xi_X(\mathrm{id}_X)$. (Construct the inverse: for $a \in \Phi(X)$, put $\xi^a_Y(f) = \Phi(f)(a)$.)

(ii) Let $U \colon \mathrm{Vect}_{\mathrm{fd}} \to \mathrm{Set}$ forget the linear structure. Show that U is represented by $\mathbb{C}$, and deduce that the natural transformations $U \Rightarrow U$ are exactly the families $v \mapsto \lambda v$, one $\lambda \in \mathbb{C}$ for all spaces at once. Show directly that $v \mapsto v + c_V$, with a nonzero vector c_V chosen in each nonzero V, is not natural.

(iii) Show that the natural transformations $\mathrm{Hom}(V, -) \Rightarrow \mathrm{Hom}(W, -)$ on $\mathrm{Vect}_{\mathrm{fd}}$ are exactly the maps $f \mapsto f \circ \psi$, one for each linear $\psi \colon W \to V$.

Hint. (i) Naturality of ξ at $f \colon X \to Y$, applied to id_X, gives $\xi_Y(f)$. (ii) Find a natural bijection $\mathrm{Hom}(\mathbb{C}, V) \to V$, then apply (i) with $\Phi = U$. (iii) Apply (i) with $\Phi = \mathrm{Hom}(W, -)$.

Answer. (i) The inverse is $a \mapsto \xi^a$. (ii) $\mathrm{Hom}(\mathbb{C}, V) \cong V$ by $f \mapsto f(1)$, so $U \Rightarrow U$ corresponds to $U(\mathbb{C}) = \mathbb{C}$; for $v + c_V$ naturality fails at $f = 0$. (iii) By (i), they correspond to $\mathrm{Hom}(W, V)$, with $\xi_Y(f) = f \circ \psi$.

Exercise 3 (The free vector space). For a set X let $\mathbb{C}[X]$ be the space of functions $X \to \mathbb{C}$ that vanish outside a finite set, with basis e_x $(x \in X)$; for a function $\alpha \colon X \to X'$ let $F(\alpha) \colon \mathbb{C}[X] \to \mathbb{C}[X']$, $e_x \mapsto e_{\alpha(x)}$. Let $U \colon \mathrm{Vect} \to \mathrm{Set}$ forget the linear structure.

(i) Show that $F\colon \mathrm{Set} \to \mathrm{Vect}$ is a functor.

(ii) Show that $F \dashv U$: $f \mapsto (x \mapsto f(e_x))$ is a bijection $\mathrm{Hom}_{\mathrm{Vect}}(\mathbb{C}[X], V) \to \mathrm{Hom}_{\mathrm{Set}}(X, U(V))$, natural in X and V. Find the unit η_X and the counit ε_V, check the two triangle identities $\varepsilon_{F(X)} \circ F(\eta_X) = \mathrm{id}_{F(X)}$ and $U(\varepsilon_V) \circ \eta_{U(V)} = \mathrm{id}_{U(V)}$, and show that the counit $\mathbb{C}[U(\mathbb{C})] \to \mathbb{C}$ is not injective.

(iii) For $|X| = 3$ and $V = \mathbb{C}^2$, what is $\dim \mathrm{Hom}_{\mathrm{Vect}}(\mathbb{C}[X], V)$?

Hint. A linear map is fixed by its values on a basis, and any values are allowed. The unit is the image of $\mathrm{id}_{\mathbb{C}[X]}$, the counit the preimage of $\mathrm{id}_{U(V)}$. For the counit on $V = \mathbb{C}$, the basis vector e_2 of $\mathbb{C}[U(\mathbb{C})]$ belongs to the number 2.

Answer. (i) $F(\beta \circ \alpha) = F(\beta) \circ F(\alpha)$ on each e_x, and $F(\mathrm{id}) = \mathrm{id}$. (ii) Unit $\eta_X(x) = e_x$; counit $\varepsilon_V(e_v) = v$; $\varepsilon_{\mathbb{C}}(e_2 - 2e_1) = 0$. (iii) 6.

Exercise 4 (The Hom-dimension table of $\mathrm{Rep}(S_3)$). In $\mathrm{Rep}(S_3)$ take the simple objects $\mathbf{1}$, sgn, std and the permutation representations $P = \mathbb{C}^3$ (on $\{1,2,3\}$), $Y = \mathbb{C}[S_3/A_3]$ (on the two cosets of A_3, swapped by the odd permutations) and $R = \mathbb{C}[S_3]$ (the regular representation).

(i) Find the multiplicities of $\mathbf{1}$, sgn, std in P, Y and R.

(ii) Fill in the 6×6 table of $\dim \mathrm{Hom}(X, X')$ for X, X' among $\mathbf{1}$, sgn, std, P, Y, R.

(iii) For which of the six is $\mathrm{End}(X)$ commutative?

Hint. A permutation representation has character $\chi(g)$ = the number of points g fixes. Take inner products with the three irreducible characters, then use $\dim \mathrm{Hom}(X, X') = \sum_i m_i(X)\, m_i(X')$ (Theorem 2.13).

Answer. (i) $P = \mathbf{1} \oplus \mathrm{std}$, $Y = \mathbf{1} \oplus \mathrm{sgn}$, $R = \mathbf{1} \oplus \mathrm{sgn} \oplus \mathrm{std}^{\oplus 2}$. (ii) In the order $\mathbf{1}$, sgn, std, P, Y, R, the rows are $(1,0,0,1,1,1)$, $(0,1,0,0,1,1)$, $(0,0,1,1,0,2)$, $(1,0,1,2,1,3)$, $(1,1,0,1,2,2)$, $(1,1,2,3,2,6)$. (iii) All but R, whose End contains $M_2(\mathbb{C})$.

Exercise 5 (Semisimple, or not). (i) Show that $\mathrm{Rep}(\mathbb{Z}_n)$ is semisimple with n simple objects, and that $\mathrm{Rep}(\mathbb{Z}_2)$ and $\mathrm{Rep}(\mathbb{Z}_3)$ are not equivalent as linear categories. You may use that an equivalence is full, faithful and essentially surjective (Leinster Prop. 1.3.18, the converse of Proposition 1.17).

(ii) An invertible 2×2 matrix A gives a representation $\rho_A(k) = A^k$ of $\mathbb{Z}$ on $\mathbb{C}^2$. Show that ρ_A is a direct sum of two one-dimensional representations exactly when A is diagonalisable. Compute $\mathrm{End}(\rho_A)$, the matrices commuting with A, for $A_1 = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, and their dimensions.

(iii) Show that $\mathrm{End}(\rho_{A_2})$ is not isomorphic, as an algebra, to any product of matrix algebras $\prod_i M_{m_i}(\mathbb{C})$. Conclude that no semisimple linear category contains an object with this endomorphism algebra.

Hint. (i) In a semisimple category, which objects X have $\mathrm{End}(X) \cong \mathbb{C}$? An equivalence is full and faithful. (ii) An invariant line is an eigenline of A. (iii) Look for a nonzero element whose square is zero in a commutative algebra.

Answer. (i) The characters $k \mapsto \zeta_n^{jk}$ are the n simples; an equivalence would match the 2 simples of $\mathrm{Rep}(\mathbb{Z}_2)$ with the 3 of $\mathrm{Rep}(\mathbb{Z}_3)$. (ii) $\mathrm{End}(\rho_{A_1}) = \left\{ \left(\begin{smallmatrix} a & b \\ 0 & a+b \end{smallmatrix}\right) \right\} \cong \mathbb{C} \times \mathbb{C}$ and $\mathrm{End}(\rho_{A_2}) = \{aI + bN\}$ with $N = A_2 - 2I$, both of dimension 2. (iii) $N \neq 0$ but $N^2 = 0$, while a commutative product of matrix algebras is $\mathbb{C}^k$.

Exercise 6 ⋆. *Frobenius reciprocity, built by hand.* Let $G = S_3$ and $H = \{1, (1\,2)\} \cong S_2$. For a representation W of H, let $\mathrm{Ind}\, W = \mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$ be the space $\mathbb{C}[G] \otimes W$ divided by the span of all $gh \otimes w - g \otimes hw$ ($g \in G$, $h \in H$, $w \in W$), with G acting by $g'(g \otimes w) = g'g \otimes w$.

(i) Let g_1, g_2, g_3 be representatives of the cosets gH. Show that $(w_1, w_2, w_3) \mapsto \sum_i g_i \otimes w_i$ is a linear isomorphism $W^3 \to \operatorname{Ind} W$, so $\dim \operatorname{Ind} W = 3 \dim W$.

(ii) On an H-intertwiner $a\colon W' \to W$ let $\operatorname{Ind}(a)(g \otimes w') = g \otimes a(w')$. Show that this is well defined, and that $f \mapsto \big(w \mapsto f(1 \otimes w)\big)$ is a bijection $\operatorname{Hom}_G(\operatorname{Ind} W, V) \to \operatorname{Hom}_H(W, \operatorname{Res} V)$, natural in W and in V: so $\operatorname{Ind} \dashv \operatorname{Res}$.

(iii) Use (ii) to decompose $\operatorname{Ind} \mathbf{1}_H$ and $\operatorname{Ind} \operatorname{sgn}_H$ into $\mathbf{1}$, sgn, std.

Hint. (i) Every g is $g_i h$ for exactly one i and h, and $g_i h \otimes w = g_i \otimes hw$ in the quotient; for the inverse, send $g \otimes w$ to hw in slot i and check that the relations go to zero. (ii) The inverse sends φ to $g \otimes w \mapsto \rho_V(g)\varphi(w)$. (iii) The multiplicity of a simple V in $\operatorname{Ind} W$ is $\dim \operatorname{Hom}_G(\operatorname{Ind} W, V)$; restrict each simple V to H.

Answer. (i) The inverse sends $g \otimes w$, with $g = g_i h$, to hw in slot i. (ii) The inverse is $\varphi \mapsto \big(g \otimes w \mapsto \rho_V(g)\varphi(w)\big)$. (iii) $\operatorname{Ind} \mathbf{1}_H = \mathbf{1} \oplus \text{std}$ and $\operatorname{Ind} \operatorname{sgn}_H = \text{sgn} \oplus \text{std}$.